

CogPilot Data Dictionary

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1. Stream: EMG

Electromyography (EMG), a measure of muscle activation, was recorded from the subject's right forearm, which controlled the flight joystick. A Shimmer Sensing device was used. EMG electrodes were placed on the forearm to capture wrist flexion and extension. To capture flexion, two electrodes were placed approximately on the brachioradialis muscle, close to the elbow. To capture extension, another two electrodes were placed approximately on the extensor digitorum, also close to the elbow. The reference electrode is placed on the bony protrusion of the ulna close to the wrist. The device also measured accelerometry (Range: $\pm 2 \text{ m/s}^2$) of the forearm.

Nominal Sampling Rate: 128 Hz, 512 Hz

Details on the EMG configuration can be found here:

https://www.shimmersensing.com/images/uploads/docs/EMG_User_Guide_Rev1.12.pdf

LabStreamingLayer (LSL) is used as the cross-modality time-synchronization mechanism. Therefore, the first variable in each dataset represents a common time across all data.

Naming convention: *sub-cp###_ses-YYYYMMDD_task-ils_stream_lslshimmeremg_feat-chunk_level-###_run-###_dat.csv*

Table 1–1: EMG Data Descriptions

Variable Name	Unit	Description
time_dn	datetime	LSL time stamps converted to MATLAB datetime format
accelerometry_forearm_r_x_mps2	m/s^2	Accelerometry of the RIGHT forearm
accelerometry_forearm_r_y_mps2	m/s^2	Accelerometry of the RIGHT forearm
accelerometry_forearm_r_z_mps2	m/s^2	Accelerometry of the RIGHT forearm
emg_wrist_flexor_mV	mV	EMG of wrist flexor muscles
emg_wrist_extensor_mV	mV	EMG of wrist extensor muscles

2. Stream: EDA

Electrodermal Activity (EDA; also known as galvanic skin response or GSR), measured on the left hand. A Shimmer Sensing device was used. The left hand was used to control the throttle. Electrodes were placed at the bases of the index and ring fingers. A photoplethysmogram (PPG) sensor was placed at the tip of the middle finger.

Nominal Sampling Rate: 128 Hz, 1024 Hz

Additional technical information can be found on the Shimmer Sensing website:

Optical Pulse Sensor:

https://www.shimmersensing.com/images/uploads/docs/Optical_Pulse_Sensor_User_Guide_rev1.6.pdf

GSR+ Sensor: https://www.shimmersensing.com/images/uploads/docs/GSR_User_Guide_rev1.13.pdf

Spec Sheet: https://www.shimmersensing.com/images/uploads/docs/GSR_Plus_Spec_Sheet_v3_4_-.pdf

Naming convention: *sub-cp###_ses-YYYYMMDD_task-ils_stream_lslshimmereda_feat-chunk_level-###_run-###_dat.csv*

Table 2–1: EDA Data Descriptions

Variable Name	Unit	Description
time_dn	datenum	LSL time stamps converted to MATLAB datenum format
ppg_finger_mV	mV	Optical PPG signal, measured at tip of the middle finger
eda_hand_l_kOhms	kOhms	EDA measured on the left hand

3. Stream: ECG

Electrocardiography (ECG) was measured using electrodes placed on the torso. A Shimmer Sensing device was used. Details can be found in the link below, but briefly, ECG electrodes were placed at the sites: Right mid-axillary line (“RA”), Left mid-axillary line (“LA”), apex of Right hip bone (“RL”; reference electrode), apex of Left hip bone (“LL”), and on the sternum below the 4th rib (“Vx”).

Nominal Sampling Rate: 128 Hz

Details on electrode placement and device specifications can be found here:

https://www.shimmersensing.com/images/uploads/docs/ECG_Respiration_User_Guide_Rev1.1.pdf

Naming convention:

sub-cp###_ses-YYYYMMDD_task-ils_stream_lslshimmerecg_feat-chunk_level-###_run-###_dat.csv

Table 3–1: ECG Data Descriptions

Variable Name	Unit	Description
time_dn	datenum	LSL time stamps converted to MATLAB datenum format
ecg_projection_ll_ra_mV	mV	ECG Signal: LL-RA projection
ecg_projection_la_ra_mV	mV	ECG Signal: LA-RA projection
ecg_projection_vx_rl_mV	mV	ECG Signal: Vx-RL projection

4. Stream: Respiration (from Shimmer Device)

Respiration is measuring using the Shimmer Sensing device. The same device used for measuring ECG (Section 3), is configured to also measure respiration. The respiration detection method used is impedance pneumography, which is a measure of the change in impedance across the chest. The impedance increases during inhalation, and decreases during exhalation.

Nominal Sampling Rate: 128 Hz

Naming convention:

sub-cp###_ses-YYYYMMDD_task-ils_stream_1slshimmerresp_feat-chunk_level-###_run-###_dat.csv

Table 4–1: Shimmer Respiration Data Descriptions

Variable Name	Unit	Description
time_dn	datetime	LSL time stamps converted to MATLAB datetime format
respiration_trace_mV	mV	Respiration trace

5. Stream: Respiration (from Resptrace Device)

Changes in the cross-sectional area of the chest wall was measured via respiratory inductive plethysmography. This was done using the Resptrace system (Ambulatory Monitoring, Ardsley, NY). A belt was fastened in the middle of the torso, around the approximate location of the diaphragm. The magnitude was zeros prior to beginning the data collection.

Nominal Sampling Rate: 1024 Hz

Naming convention:

sub-cp###_ses-YYYYMMDD_task-ils_stream_1sresptrace_feat-chunk_level-###_run-###_dat.csv

Table 5–1: Resptrace Data Descriptions

Variable Name	Unit	Description
time_dn	datenum	LSL time stamps converted to MATLAB datenum format
respiration_trace_v	volts	Respiration trace

6. Stream: ACC

Three-axis accelerometry (ACC), measured on the ventral aspect of the torso of the volunteer. A Shimmer Sensing device was strapped to the torso adjacent to the device that was used to measure ECG and Respiration. The device was positioned approximately at or right above the beltline.

Nominal Sampling Rate: 128 Hz

Naming convention:

sub-cp###_ses-YYYYMMDD_task-ils_stream_1slshimmertorsoacc_feat-chunk_level-###_run-###_dat.csv

Table 6–1: Torso Accelerometry Descriptions

Variable Name	Unit	Description
time_dn	datenum	LSL time stamps converted to MATLAB datenum format
accelerometry_torso_x_mps2	m/s^2	Accelerometry of the TORSO
accelerometry_torso_y_mps2	m/s^2	Accelerometry of the TORSO
accelerometry_torso_z_mps2	m/s^2	Accelerometry of the TORSO

7. Stream: HTC Vive Eye

Eye tracking data streamed from the HTC Vive Pro Eye. Left and Right eye were tracked independently. Using the HTC Vive in-built functionality, an eye tracking calibration was performed prior to the start of the experiment and after anytime the headset was adjusted or removed.

Nominal Sampling Rate: 250 Hz

Information on the HTC Vive SDK used to acquire the data can be found here, and associated webpages:

<https://developer.vive.com/resources/vive-sense/hardware-guide/vive-pro-eye-specs-user-guide/>

The SDK, with documentation on variables, can be downloaded as a zip file from:

<https://hub.vive.com/en-US/download>

Naming convention: *sub-cp###_ses-YYYYMMDD_task-ils_stream_lshtcviveye_feat-chunk_level-###_run-###_dat.csv*

Table 7–1: HTC Vive Eye Data Descriptions

Variable Name	Unit	Description
time_dn	datenum	LSL time stamps converted to MATLAB datenum format
validity_l	-	The bits containing all validity for this frame - LEFT eye
validity_r	-	The bits containing all validity for this frame – RIGHT eye
gaze_origin_l_x_mm	mm	The point in the LEFT eye from which the gaze ray originates (right-handed coordinate system)
gaze_origin_l_y_mm	mm	The point in the LEFT eye from which the gaze ray originates (right-handed coordinate system)
gaze_origin_l_z_mm	mm	The point in the LEFT eye from which the gaze ray originates (right-handed coordinate system)
gaze_origin_r_x_mm	mm	The point in the RIGHT eye from which the gaze ray originates (right-handed coordinate system)
gaze_origin_r_y_mm	mm	The point in the RIGHT eye from which the gaze ray originates (right-handed coordinate system)
gaze_origin_r_z_mm	mm	The point in the RIGHT eye from which the gaze ray originates (right-handed coordinate system)
gaze_direction_l_x	-	The normalized gaze direction of the LEFT eye in [0,1] (right-handed coordinate system)
gaze_direction_l_y	-	The normalized gaze direction of the LEFT eye in [0,1] (right-handed coordinate system)
gaze_direction_l_z	-	The normalized gaze direction of the LEFT eye in [0,1] (right-handed coordinate system)
gaze_direction_r_x	-	The normalized gaze direction of the RIGHT eye in [0,1] (right-handed coordinate system)
gaze_direction_r_y	-	The normalized gaze direction of the RIGHT eye in [0,1] (right-handed coordinate system)

Variable Name	Unit	Description
gaze_direction_r_z	-	The normalized gaze direction of the RIGHT eye in [0,1] (right-handed coordinate system)
pupil_diameter_l_mm	mm	Diameter of the LEFT pupil
pupil_diameter_r_mm	mm	Diameter of the RIGHT pupil
eye_openness_l	-	A value representing how open the LEFT eye is
eye_openness_r	-	A value representing how open the RIGHT eye is
pupil_position_l_x	-	Normalized position of LEFT pupil in sensor area
pupil_position_l_y	-	Normalized position of LEFT pupil in sensor area
pupil_position_l_z	-	Normalized position of LEFT pupil in sensor area
pupil_position_r_x	-	Normalized position of RIGHT pupil in sensor area
pupil_position_r_y	-	Normalized position of RIGHT pupil in sensor area
pupil_position_r_z	-	Normalized position of RIGHT pupil in sensor area
convergence_distance_mm	mm	Combined eye measure of depth of binocular fixation
convergence_distance_validity	-	Boolean report of convergence data validity

8. Stream: X-Plane

Aircraft and pilot state data streamed from X-Plane 11. Pilot state data can be used to translate eye tracking data. Aircraft state data can be used to compute flight performance metrics.

Nominal Sampling Rate: 4 Hz

More information on the X-Plane Datarefs can be found here: <https://developer.x-plane.com/datarefs/>
(Note: Dataref descriptions on this site may contain incorrect units or misleading information; correct descriptions for a subset of fields is provided below.)

Naming convention: *sub-cp###_ses-YYYYMMDD_task-ils_stream_lslxp11xpcplt_feat-chunk_level-###_run-###_dat.csv*

Table 8–1: X-Plane Pilot Data Descriptions

Variable Name	Unit	Description
time_dn	datenum	LSL time stamps converted to MATLAB datenum format
pilot_head_position_h_m	m	Pilot head horizontal position (positive is right)
pilot_head_position_f_m	m	Pilot head forward position (positive is back)
pilot_head_position_v_m	m	Pilot head vertical position (positive is up)
pilot_head_yaw_deg	deg	Pilot head yaw angle (positive is right)
pilot_head_pitch_deg	deg	Pilot head pitch angle (positive is up)
pilot_head_roll_deg	deg	Pilot head roll angle

Naming convention: *sub-cp###_ses-YYYYMMDD_task-ils_stream_lslxp11xpcac_feat-chunk_level-###_run-###_dat.csv*

Table 8–2: X-Plane Aircraft Data Descriptions

Variable Name	Unit	Description
time_dn	datenum	LSL time stamps converted to MATLAB datenum format
aircraft_latitude_deg	deg	Aircraft latitude
aircraft_longitude_deg	deg	Aircraft longitude
aircraft_elevation_m	m	Aircraft height above WGS84 ellipsoid
aircraft_agl_altitude_m	m	Aircraft altitude above ground level (AGL)
aircraft_indicated_airspeed_kias	kias	Aircraft air speed indicated
aircraft_groundspeed_mps	m/s	Aircraft ground speed
aircraft_velocity_e_mps	m/s	Aircraft speed component along true east vector
aircraft_velocity_u_mps	m/s	Aircraft inertial vertical speed component pointing up
aircraft_velocity_n_mps	m/s	Aircraft speed component along true north vector
aircraft_climb_rate_mps	m/s	Aircraft climb rate
aircraft_landing_gear	pct	Aircraft landing gear deployment state

Variable Name	Unit	Description
aircraft_ils_deflection_h	pct	Aircraft instrument landing system (ILS) horizontal deflection
aircraft_ils_deflection_gs	pct	Aircraft instrument landing system (ILS) glide slope deflection
aircraft_yaw_deg	deg	Aircraft yaw angle (positive is right)
aircraft_pitch_deg	deg	Aircraft pitch angle (positive is up)
aircraft_roll_deg	deg	Aircraft roll angle (positive is left / counter-clockwise)
aircraft_elevation_trim	n/a	Elevation Trim, -1 = max nose down, 1 = max nose up
aircraft_aileron_trim	n/a	Aileron Trim, -1 = max left, 1 = max right

9. Performance Metrics

The ideal descent involves minimizing deviation away from a straight path towards the runway as well as maintaining the optimal speed of 115 knots. As such, the performance of a flight was determined based on the three components of Localizer (horizontal) error, Glide Slope (vertical) error, and airspeed error. The instantaneous angular deviations of Localizer and Glide Slope error were derived based on aircraft latitude, longitude, and elevation relative to the runway. The airspeed error was derived based on the aircraft indicated airspeed. These signals were obtained from the X-Plane data stream, the component errors were computed, scaled, and restricted to a range of -3 to $+3$.

Flight errors were only computed between start of flight (speed > 0) and before touchdown (AGL $> 200'$). Below the 200' altitude would be considered “minimums” and trained pilots would visually land the plane, rather than use the instruments.

Nominal Sampling Rate: 4 Hz

Naming convention: *sub-cp###_ses-YYYYMMDD_task-ils_stream_lslxp11_feat-perfmetric_run-###.csv*

Table 9–1: Performance Metrics Data Descriptions

Variable Name	Unit	Description
time_dn	datenum	LSL time stamps converted to MATLAB datenum format
localizer_error_deg	deg	Derived from horizontal angular deviation from the center of the runway.
glideslope_error_deg	deg	Derived from vertical deviation away from a -3 deg descent to the runway; capped at ± 3 deg.
airspeed_error_kts	knots	Derived from deviation away from 115 knots; divided by 10 and capped at ± 3 .
total_error	-	Instantaneous sum of the magnitude of the localizer, glide slope, and airspeed errors

As a convenient aggregation of error across all subjects, a summary of the flight performance error is included. The summary file includes the cumulative error for each of the three components, as well as the cumulative total error for that trial, for each subject and trial. The cumulative errors are calculated as the sum of the magnitude of the instantaneous errors across the whole trial.

Naming convention: *dataPackage/task-ils/PerfMetrics.csv*

Table 9–2: Summary Table Data Descriptions

Variable Name	Unit	Description
subject	-	Subject ID
date	-	Date of experimental session
run	-	Trial number within experiment
difficulty	-	Difficulty level of given trial (range: 1 – 4)
cumulative_localizer_error_deg	deg	Sum of the magnitude of <i>localizer_error</i> for that entire trial
cumulative_glideslope_error_deg	deg	Sum of the magnitude of the <i>glideslope_error</i> for that entire trial

cumulative_airspeed_error_kts	knots	Sum of the magnitude of the <i>airspeed_error</i> for that entire trial
cumulative_total_error	-	Sum of the <i>total_error</i> for that entire trial